

Abstract

Authorship Verification (AV) is a fundamental binary NLU task that determines whether two distinct texts share the same author. It plays a critical role in forensic linguistics, plagiarism detection, and combating digital misinformation. In this study, we tackle the inherent difficulties of cross-topic authorship attribution by presenting Group 33's two complementary approaches:

Model A: Stylometric Feature Ensemble

High-efficiency pipeline with no neural components.

Macro F1 = **0.7702**

Model C: RoBERTa-Large + Asymmetric Loss (ASL)

Robust fine-tuning to handle mixed-domain overfitting.

Macro F1 = **0.8348** (+0.0646 over Model A)

Introduction

Problem Definition: Given a pair of texts (s_1, s_2), the objective is to predict $y=1$ (same author) or $y=0$ (different author). This requires models to effectively isolate an author's intrinsic *stylistic signature* from the semantic topic.

Key Challenges in Real-World AV

- **Stylistic Sparsity:** Short, informal texts provide highly limited lexical and syntactic cues.
- **Domain & Genre Shifting:** An author's writing style naturally fluctuates across contexts (e.g., email vs. chat).
- **Intra-Author Variation:** Individuals exhibit stylistic inconsistencies over time, making static profiling difficult.
- **Closed-Track Restriction:** No external stylistic datasets are permitted, strictly limiting the learning boundary.

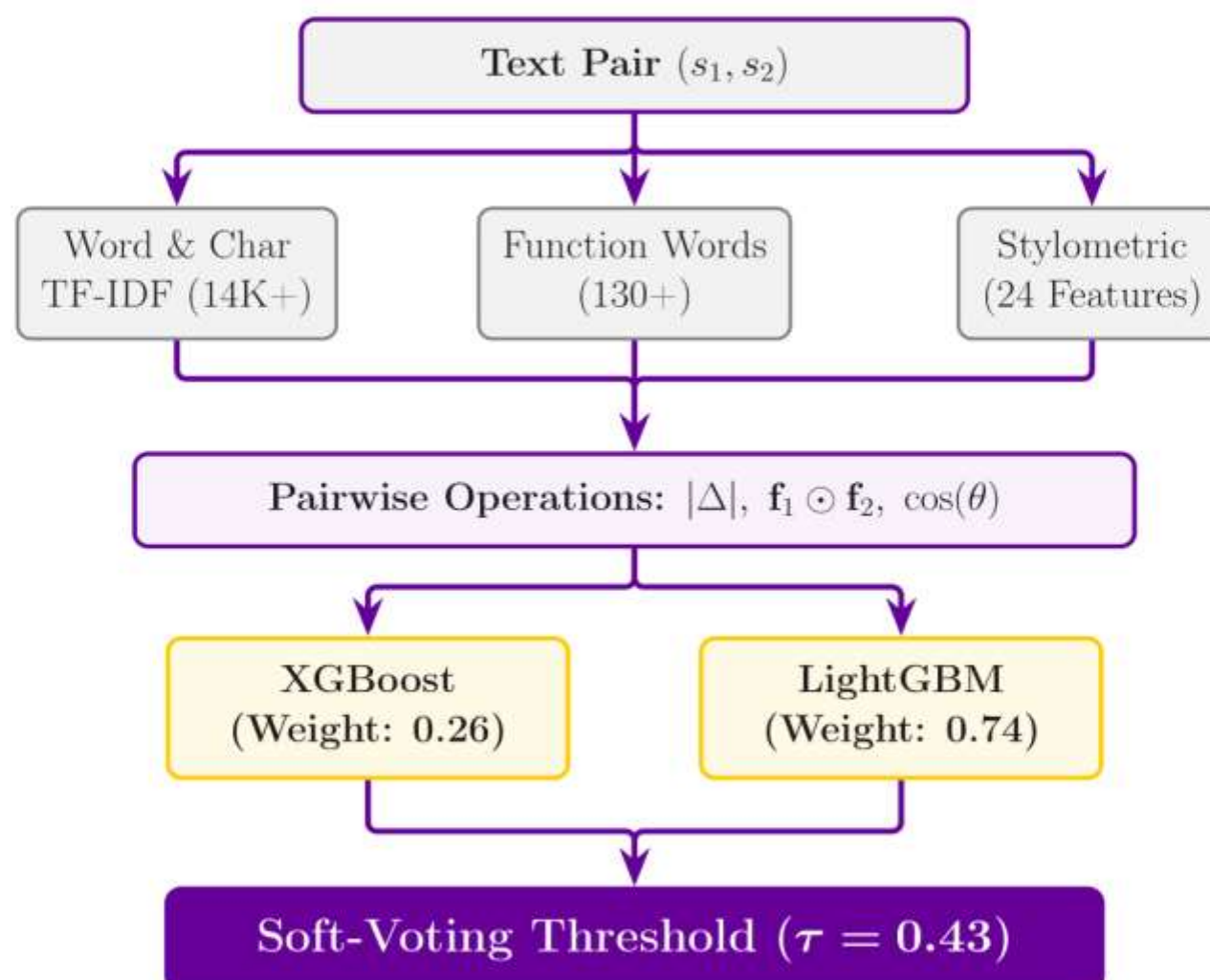
Dataset Overview

The project utilizes a **closed-track** dataset provided for the COMP34812 AV task. It comprises a highly diverse set of real-world texts, primarily drawn from corporate communications, reflecting various lengths and linguistic styles.

Split	Pairs	Note
Train	27,000+	Unlabelled; labels provided separately
Validation	5,993	Dev F1 eval. (Label 0/1 $\approx$ 50%)
Test	Hidden	Final submission track

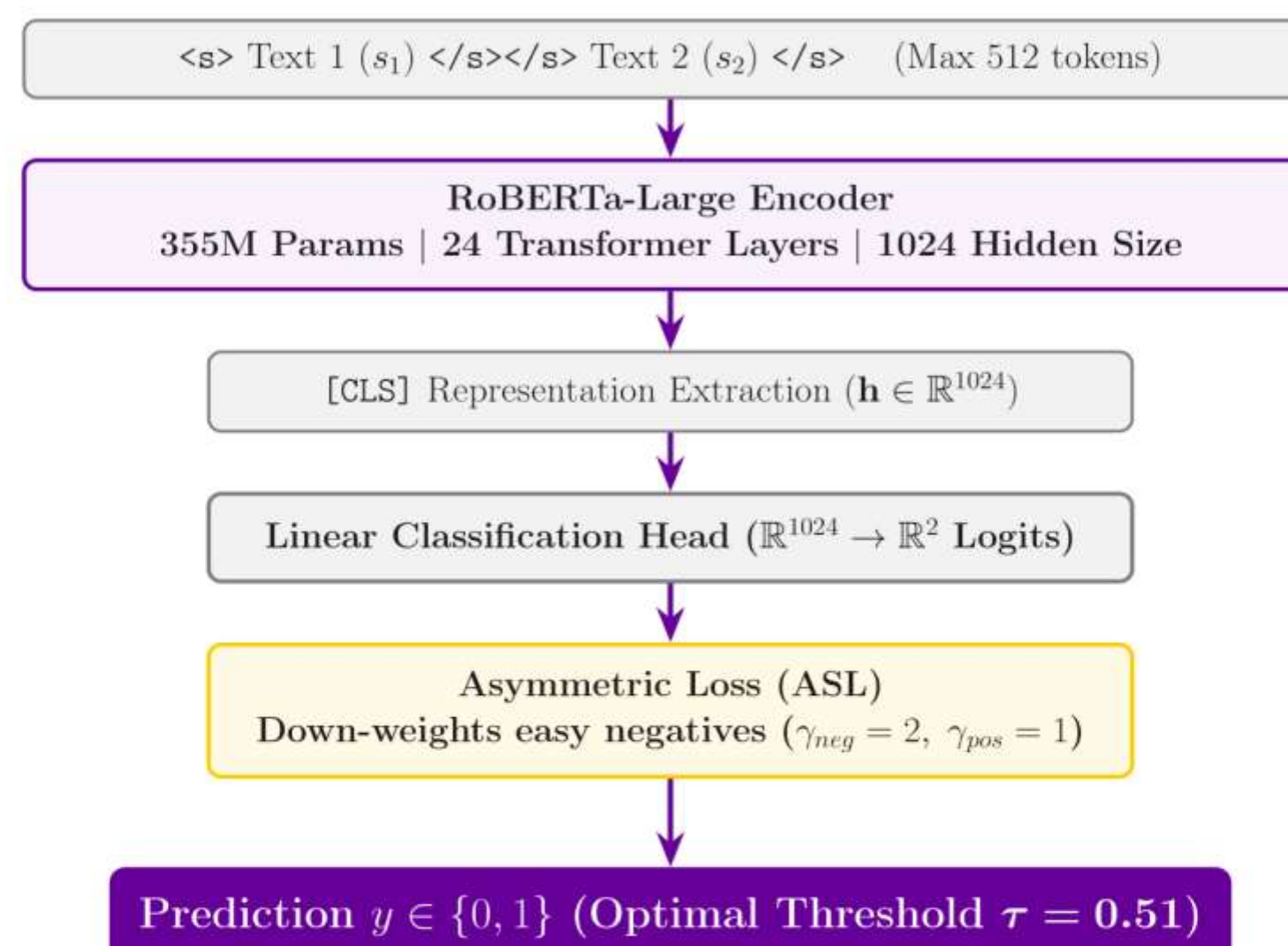
Model A: Stylometric Ensemble

Model A takes a highly engineered, feature-driven Machine Learning approach. By extracting over 14,000 multidimensional stylometric signals, it leverages a robust soft-voting ensemble of gradient boosting models (XGBoost & LightGBM). This pipeline ensures extreme computational efficiency while effectively capturing intricate lexical and syntactic nuances between text pairs.



Model C: RoBERTa-Large & ASL

Model C leverages the immense representational capacity of a 355M-parameter **RoBERTa-Large** encoder to extract deep, context-aware stylistic features. Our core architectural innovation is the integration of **Asymmetric Loss (ASL)**. By dynamically decoupling the penalty for positive and negative classes, ASL aggressively down-weights easily distinguishable negatives. This effectively neutralizes severe domain overfitting and forces the model to focus on the most challenging, ambiguous text pairs.

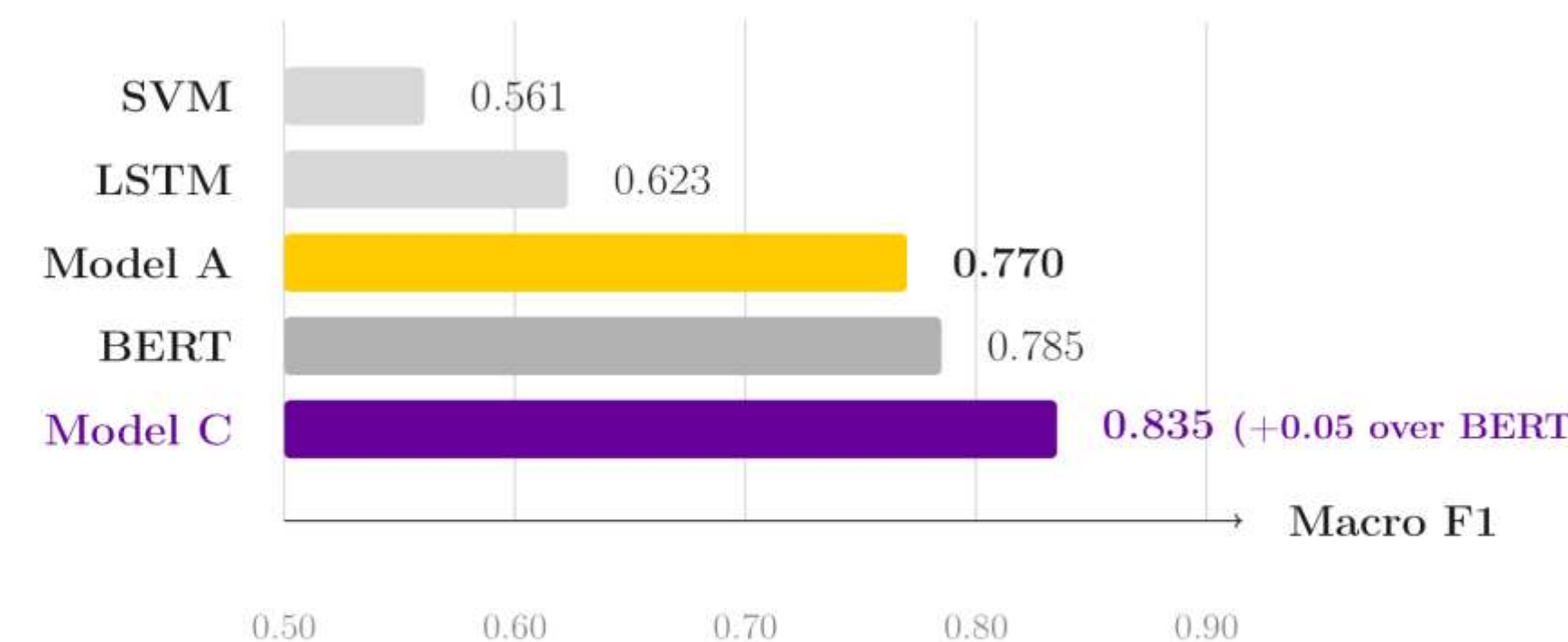


Why ASL over Cross-Entropy (CE)?

CE caused rapid overfitting on easy negatives, leading to an unstable threshold (0.38) and high validation loss (1.12). ASL dynamically down-weights easy examples, yielding a highly stable learning curve (Val Loss 0.27) and a balanced decision threshold ($\tau = 0.51$).

Experimental Results

Model C overwhelmingly surpasses all official baselines, demonstrating the superiority of the ASL-fine-tuned RoBERTa-Large architecture. Model A also shows highly competitive performance given its computational efficiency.



Discussion & Conclusion

Discussion on Model Trade-offs & Limitations:

While Model C achieves state-of-the-art accuracy, RoBERTa's absolute positional embeddings restrict inputs to 512 tokens. For heavily concatenated threads or extensive documents, trailing stylistic cues are inevitably truncated. Here, Model A reveals its strategic value: its stylometric extraction natively bypasses length constraints, offering a highly interpretable, white-box alternative that captures global document statistics without truncation.

Conclusion:

This project successfully demonstrates a holistic approach to Authorship Verification. By bridging traditional stylometry (Model A) and deep semantic modeling (Model C), we address both length limitations and complex stylistic nuances. Crucially, integrating **Asymmetric Loss (ASL)** into the Transformer architecture effectively suppressed domain-shift overfitting on easy negatives, propelling our model to fundamentally outperform standard baselines.

Ablation Insights

Ablation Study: Why not DeBERTa-v3?

Extensive ablation studies were conducted utilizing modern architectures. Surprisingly, the highly capable **DeBERTa-v3-base** yielded sub-optimal performance (Max F1 $\approx$ 0.77), falling short of even the standard BERT model. We hypothesize this degradation stems from:

- **Objective Clash:** DeBERTa's RTD (Replaced Token Detection) pre-training conflicted with the nuanced stylistic signals required for AV, particularly under the ASL framework.
- **Domain Mismatch:** DeBERTa's pre-training corpus aligns less effectively with the idiosyncratic structures of the Enron email dataset compared to RoBERTa's robust byte-level BPE.

Reproducibility: All models and training logs are available at github.com/LeonVir/NLU